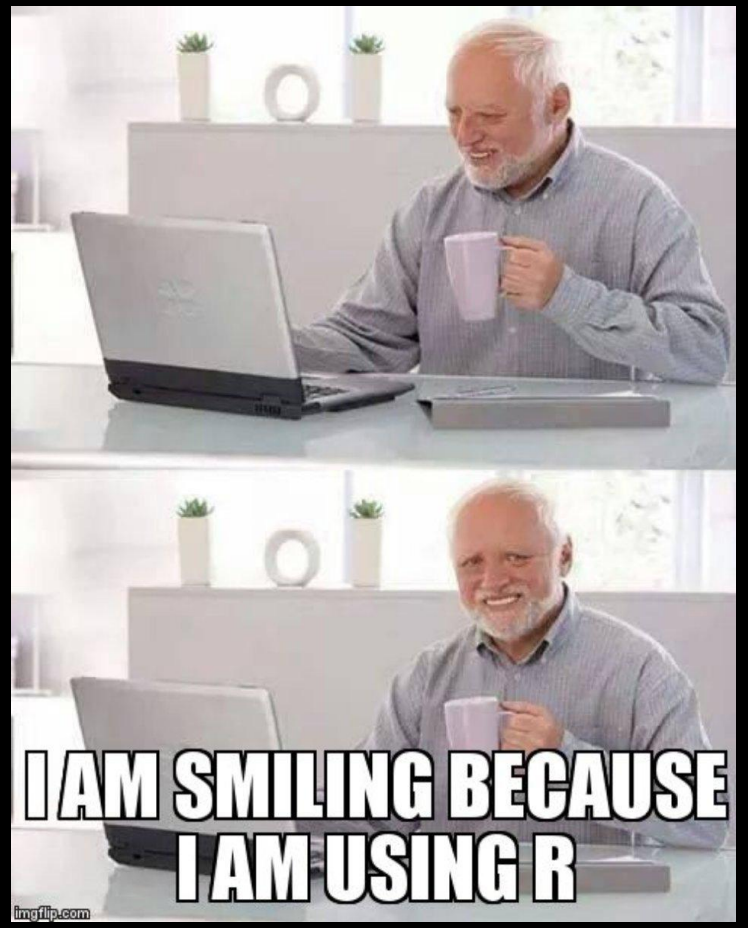


MES 503 Lab 6

Kayla Blincow, PhD

Announcements

- Lab assignment 2 due today

TENTATIVE SCHEDULE OF TOPICS – MES 503 Fall 2026

		Mondays (Lectures)		Wednesdays (Labs)	
Week	**Date**	**Lecture Topic**	**Date**	**Lab topic**	**Assessments**
1	8/17	Introduction to Statistics & Experimental Design I	8/19	Introduction to R and RStudio	
2	8/24	Introduction to Statistics & Experimental Design II	8/26	Intro to ggplot2	
3	8/31	Correlation	9/2	Correlation in R	Quiz 1
4	9/7	LABOR DAY (No Class)	9/9	Intro to Data wrangling	Lab 1 Due (9/10)
5	9/14	Statistical Hypothesis Testing & t tests	9/16	t Tests in R	
6	9/21	Linear Regression	9/23	Linear Regression in R	Quiz 2 (9/21) Lab 2 Due (9/24)
7	9/28	ANOVA	9/30	ANOVA in R	
8	10/5	ANOVA II (Two-Way & Nested) (virtual)	10/7	Two-Way & Nested ANOVA in R	Quiz 3
9	10/12	Multiple Linear Regression	10/14	Multiple Linear Regression in R	Lab 3 Due (10/15)
10	10/19	ANCOVA	10/21	ANCOVA in R	Quiz 4
11	10/26	Analysis of Frequencies	10/28	Analysis of Frequencies in R	Lab 4 Due (10/29)
12	11/2	Liberty Day – UVI Closed	11/4	Miscellaneous Fun R Topic	Quiz 5
13	11/9	Quantifying Variability/Figure and Data Sheet Design	11/11	Veterans Day – UVI Closed	
14	11/16	Reading for Stats Presentations	11/18	Reading for Stats Presentations	Lab 5 Due (11/17)
15	11/23	Applying this course to your thesis	11/25	Study Day	
16			12/3	**Final Exam**	


ACTIVITY TIME
makeameme.org

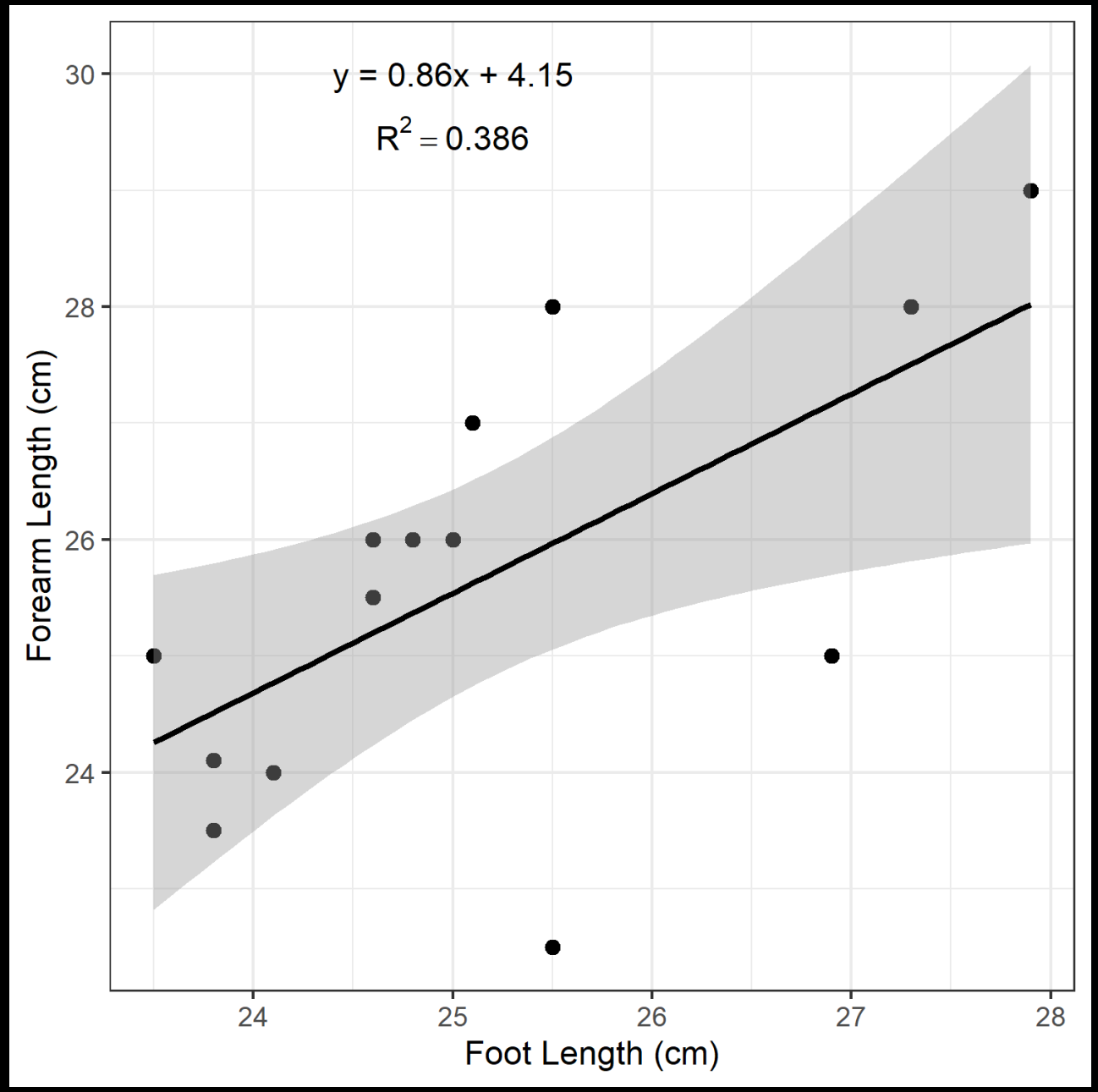
Shoe Size Chart - Women's Sizes

US Sizes	Euro Sizes	UK Sizes	Inches	CM
4	35	2	8.1875"	20.8
4.5	35	2.5	8.375"	21.3
5	35-36	3	8.5"	21.6
5.5	36	3.5	8.75"	22.2
6	36-37	4	8.875"	22.5
6.5	37	4.5	9.0625"	23
7	37-38	5	9.25"	23.5
7.5	38	5.5	9.375"	23.8
8	38-39	6	9.5"	24.1
8.5	39	6.5	9.6875"	24.6
9	39-40	7	9.875"	25.1
9.5	40	7.5	10"	25.4
10	40-41	8	10.1875"	25.9
10.5	41	8.5	10.3125"	26.2
11	41-42	9	10.5"	26.7
11.5	42	9.5	10.6875"	27.1
12	42-43	10	10.875"	27.6

TOP

Shoe Size Chart - Men's Sizes

US Sizes	Euro Sizes	UK Sizes	Inches	CM
6	39	5.5	9.25"	23.5
6.5	39	6	9.5"	24.1
7	40	6.5	9.625"	24.4
7.5	40-41	7	9.75"	24.8
8	41	7.5	9.9375"	25.4
8.5	41-42	8	10.125"	25.7
9	42	8.5	10.25"	26
9.5	42-43	9	10.4375"	26.7
10	43	9.5	10.5625"	27
10.5	43-44	10	10.75"	27.3
11	44	10.5	10.9375"	27.9
11.5	44-45	11	11.125"	28.3
12	45	11.5	11.25"	28.6
13	46	12.5	11.5625"	29.4
14	47	13.5	11.875"	30.2
15	48	14.5	12.1875"	31
16	49	15.5	12.5"	31.8



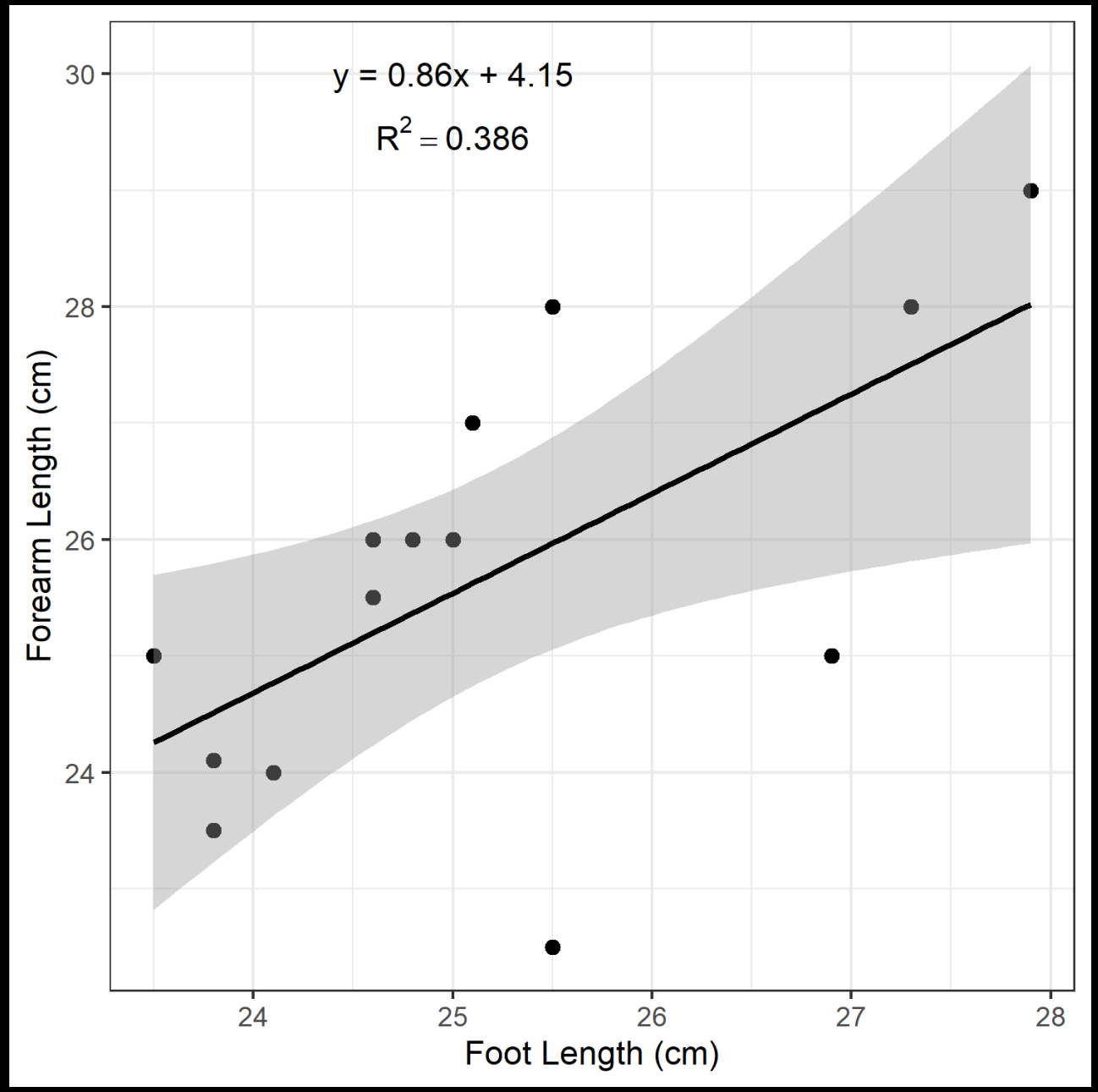

```
> summary(m1)

Call:
lm(formula = Forearm.cm. ~ FootSize.cm., data = armfoot)

Residuals:
    Min      1Q  Median      3Q     Max
-3.4668 -0.6799  0.4771  0.7885  2.0332

Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept)    4.1499     7.8537   0.528   0.6069
FootSize.cm.   0.8556     0.3116   2.746   0.0177 *
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 1.517 on 12 degrees of freedom
Multiple R-squared:  0.3859,    Adjusted R-squared:  0.3347
F-statistic: 7.539 on 1 and 12 DF,  p-value: 0.01774
```



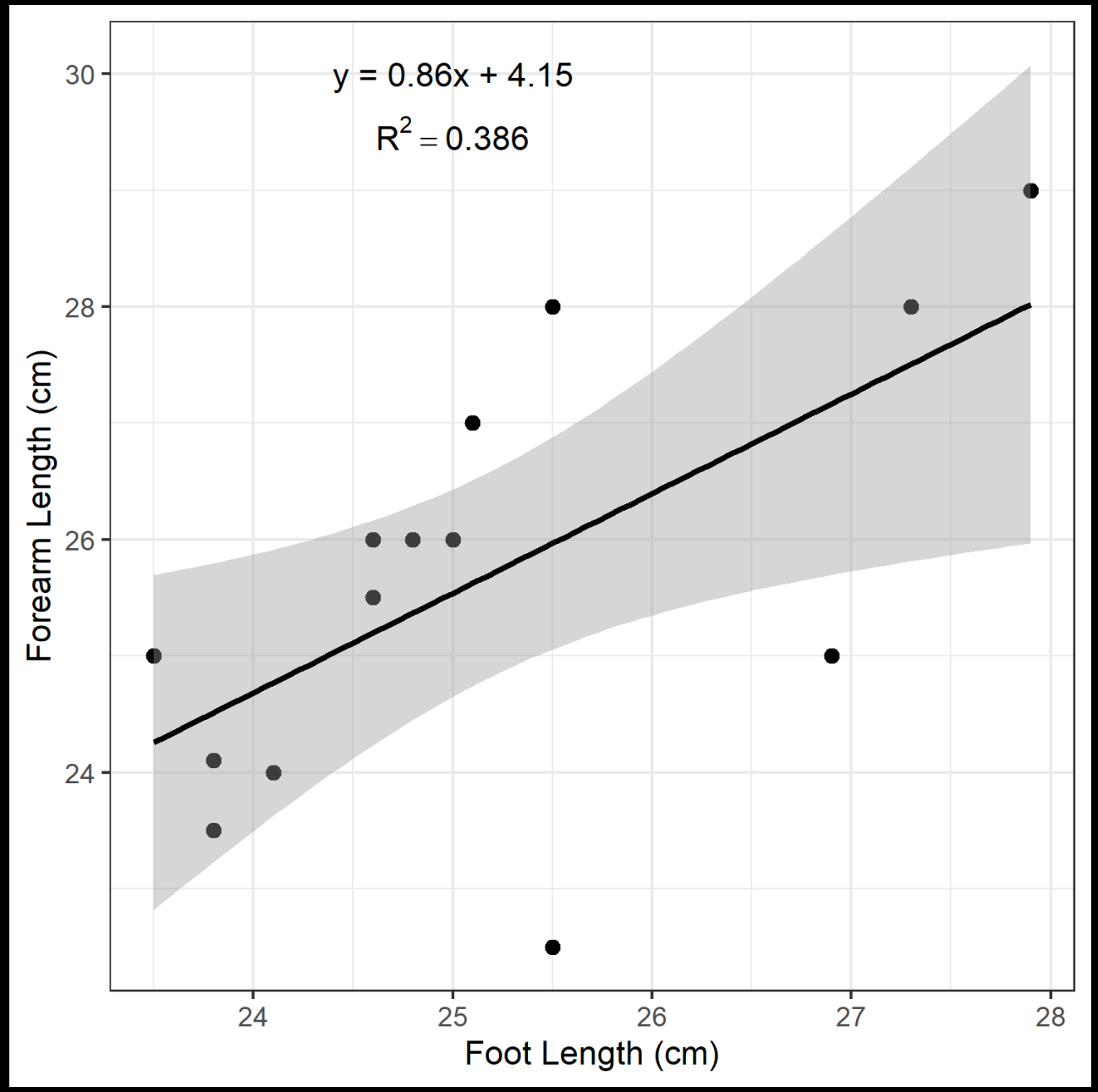

```
> summary(m1)

Call:
lm(formula = Forearm.cm. ~ FootSize.cm., data = armfoot)

Residuals:
    Min      1Q  Median      3Q     Max
-3.4668 -0.6799  0.4771  0.7885  2.0332

Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept)    4.1499     7.8537   0.528   0.6069
FootSize.cm.   0.8556     0.3116   2.746   0.0177 *
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 1.517 on 12 degrees of freedom
Multiple R-squared:  0.3859,    Adjusted R-squared:  0.3347
F-statistic: 7.539 on 1 and 12 DF,  p-value: 0.01774
```



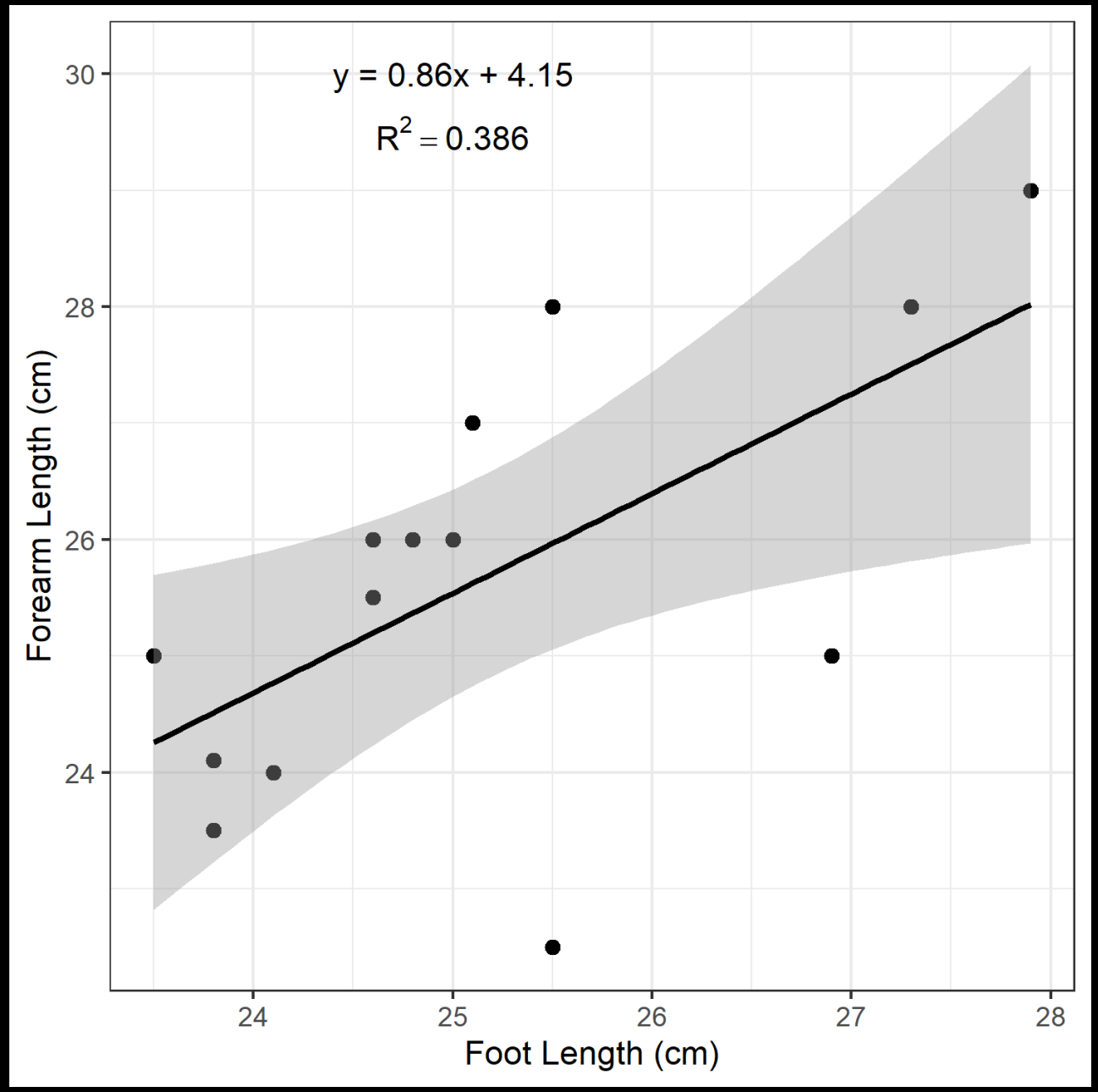

```
> summary(m1)

Call:
lm(formula = Forearm.cm. ~ FootSize.cm., data = armfoot)

Residuals:
    Min      1Q  Median      3Q     Max
-3.4668 -0.6799  0.4771  0.7885  2.0332

Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept)    4.1499     7.8537   0.528   0.6069
FootSize.cm.   0.8556     0.3116   2.746   0.0177 *
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 1.517 on 12 degrees of freedom
Multiple R-squared:  0.3859,    Adjusted R-squared:  0.3347
F-statistic: 7.539 on 1 and 12 DF,  p-value: 0.01774
```



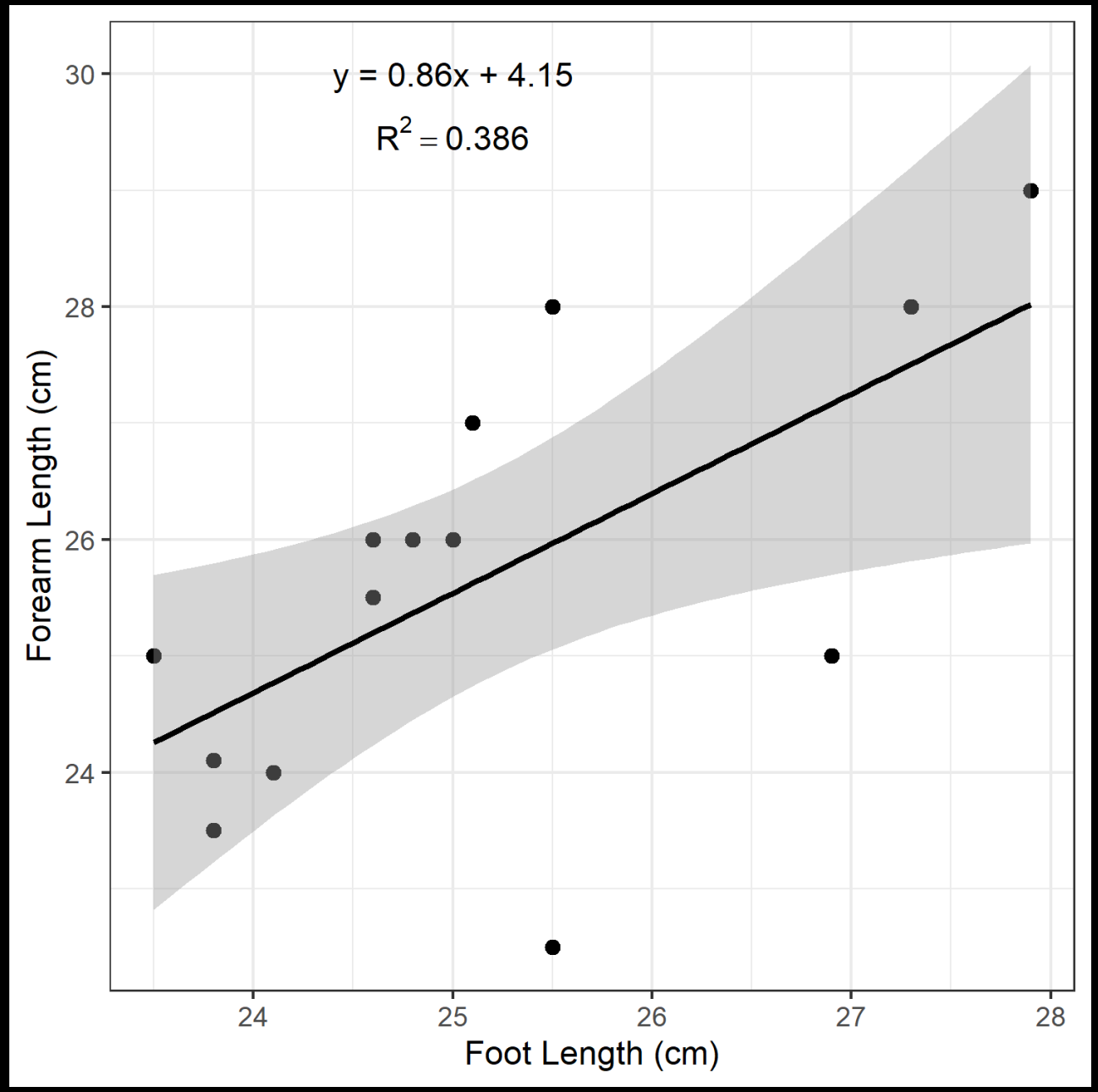

```
> summary(m1)

Call:
lm(formula = Forearm.cm. ~ FootSize.cm., data = armfoot)

Residuals:
    Min      1Q  Median      3Q     Max
-3.4668 -0.6799  0.4771  0.7885  2.0332

Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept)    4.1499     7.8537   0.528   0.6069
FootSize.cm.   0.8556     0.3116   2.746   0.0177 *
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 1.517 on 12 degrees of freedom
Multiple R-squared:  0.3859,    Adjusted R-squared:  0.3347
F-statistic: 7.539 on 1 and 12 DF,  p-value: 0.01774
```




```
> summary(m1)

Call:
lm(formula = Forearm.cm. ~ FootSize.cm., data = armfoot)

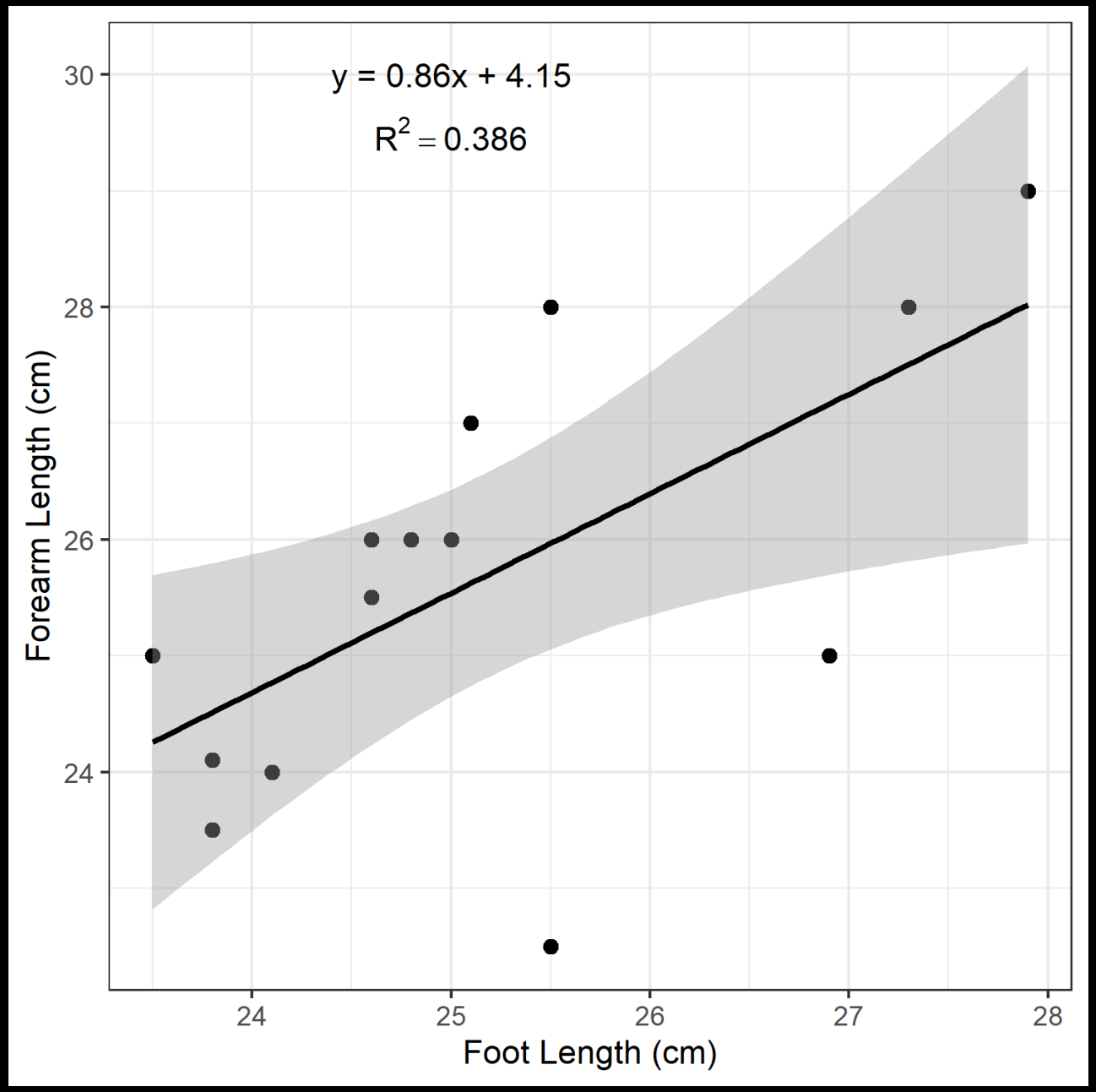
Residuals:
    Min      1Q  Median      3Q     Max
-3.4668 -0.6799  0.4771  0.7885  2.0332

Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept)    4.1499     7.8537   0.528   0.6069
FootSize.cm.   0.8556     0.3116   2.746   0.0177 *
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 1.517 on 12 degrees of freedom
Multiple R-squared:  0.3859,    Adjusted R-squared:  0.3347
F-statistic: 7.539 on 1 and 12 DF,  p-value: 0.01774
```

Reporting Regression Results

- There was a significant relationship between [independent variable] and [dependent variable] (R2 = [R2 value], F(df regression, df residual) = [F-value], p = [p-value]).
- [independent variable] significantly predicted [dependent variable] (β = [β-value], p = [p-value]).



There was a significant relationship between the length of foot and the length of the forearm (R2 = 0.386, F(1, 12) = 7.539, p = 0.018. The length of the foot significantly predicted the length of the forearm (β = 0.86, p = 0.018).

Switch to ze code!

TODAY!
More Giant Sea Bass

Importance of Trophic Ecology

- Understand how recovery might impact ecosystem dynamics
  - Predators can have disproportionate influence (e.g. Donohue et al. 2017)



Stable Isotope Analysis

Bulk Stable Isotopes







^{13}C

^{15}N



$\delta^{15}N$





$\delta^{13}C$

Stable Isotope Analysis

Bulk Stable Isotopes







^{13}C

^{15}N

$\delta^{15}N$









$\delta^{13}C$

Stable Isotope Analysis

Bulk Stable Isotopes







^{13}C

^{15}N











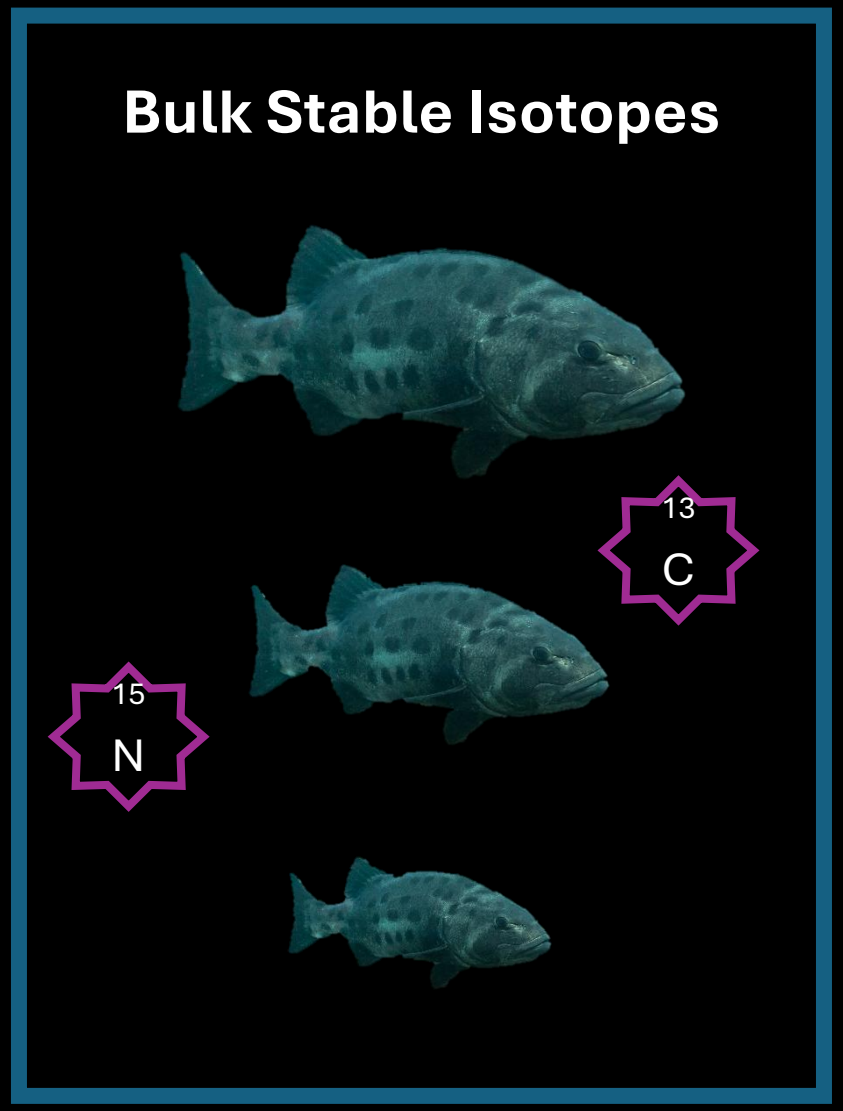


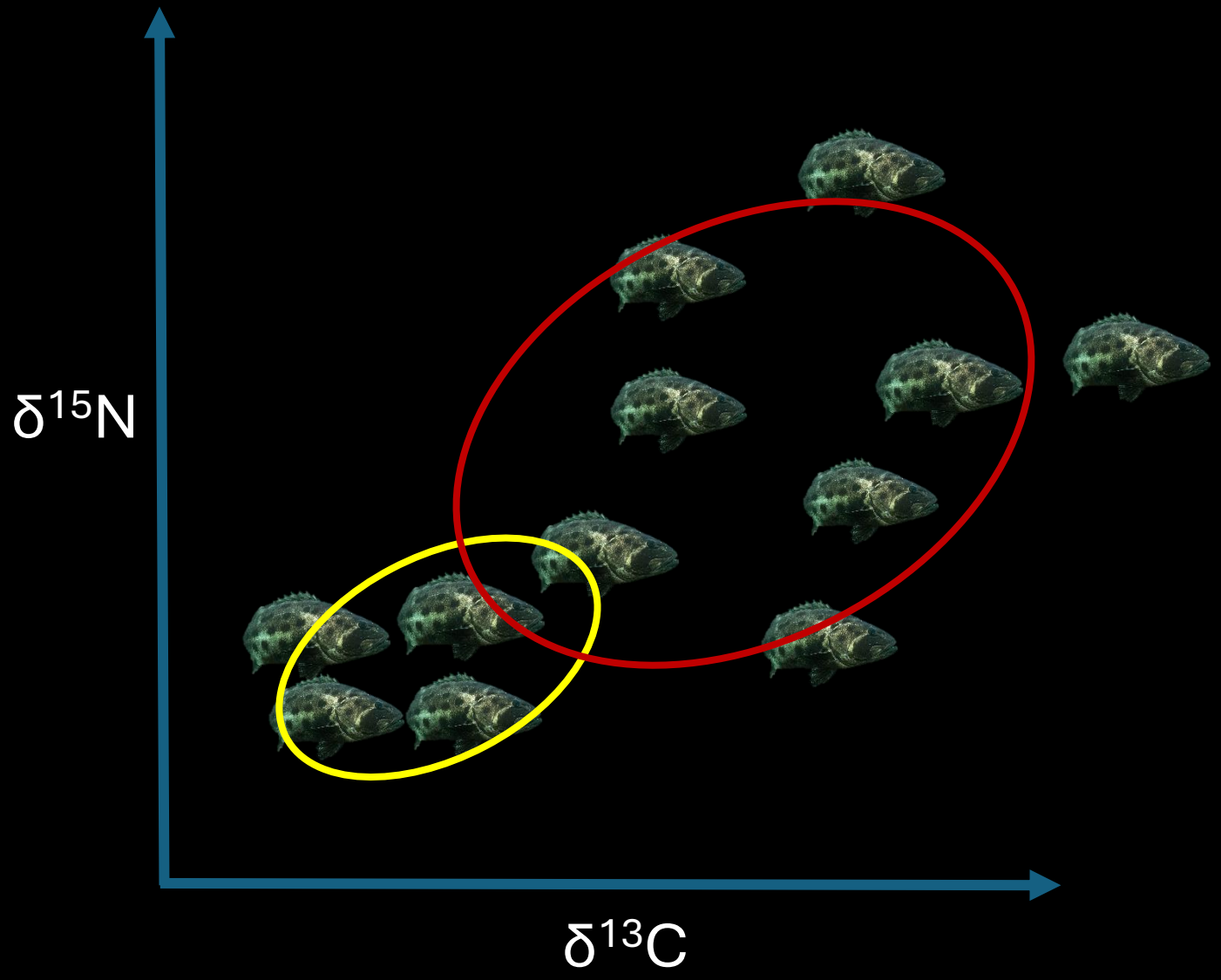


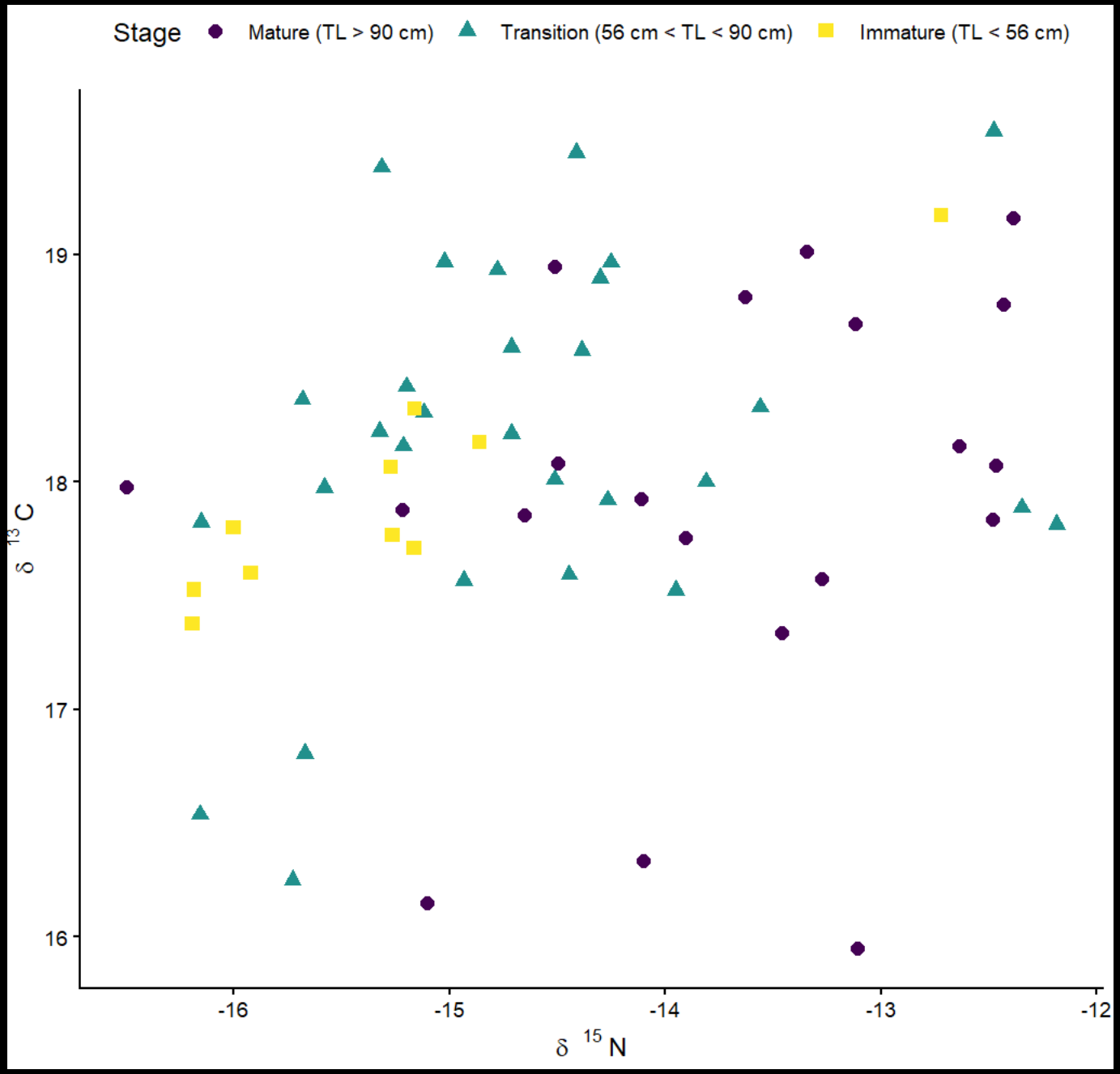
$\delta^{15}N$

$\delta^{13}C$

Stable Isotope Analysis






Stage
Mature (TL > 90 cm)
Transition (56 cm < TL < 90 cm)
Immature (TL < 56 cm)
19
18
17
16
δ 13 C
-16
-15
-14
-13
-12
δ 15 N

Extra Resources

- How to Interpret P-values and Coefficients in Regression Analysis - Statistics By Jim
- The Main Ideas of Fitting a Line to Data (The Main Ideas of Least Squares and Linear Regression.) - YouTube
- Linear Regression, Clearly Explained!!! – YouTube
- What are degrees of freedom in statistics? A simple explanation. - YouTube